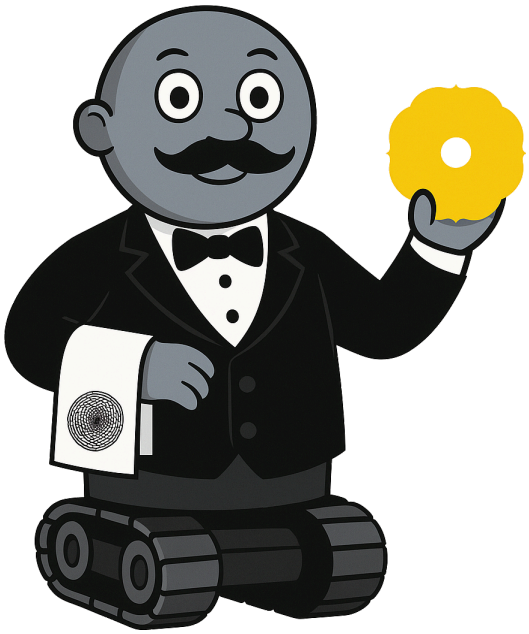


Rose Engine Butler

System Installation and Setup for Debian 13 Trixie

Version 0.19
22 January 2026

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Overview

This document is for installing and setting up LinuxCNC on the Raspberry Pi 5 with a Mesa FPGA board.

Prerequisites

This set of instructions requires you to have the following:

To get the software downloaded for installing and setting up REB, you will need:

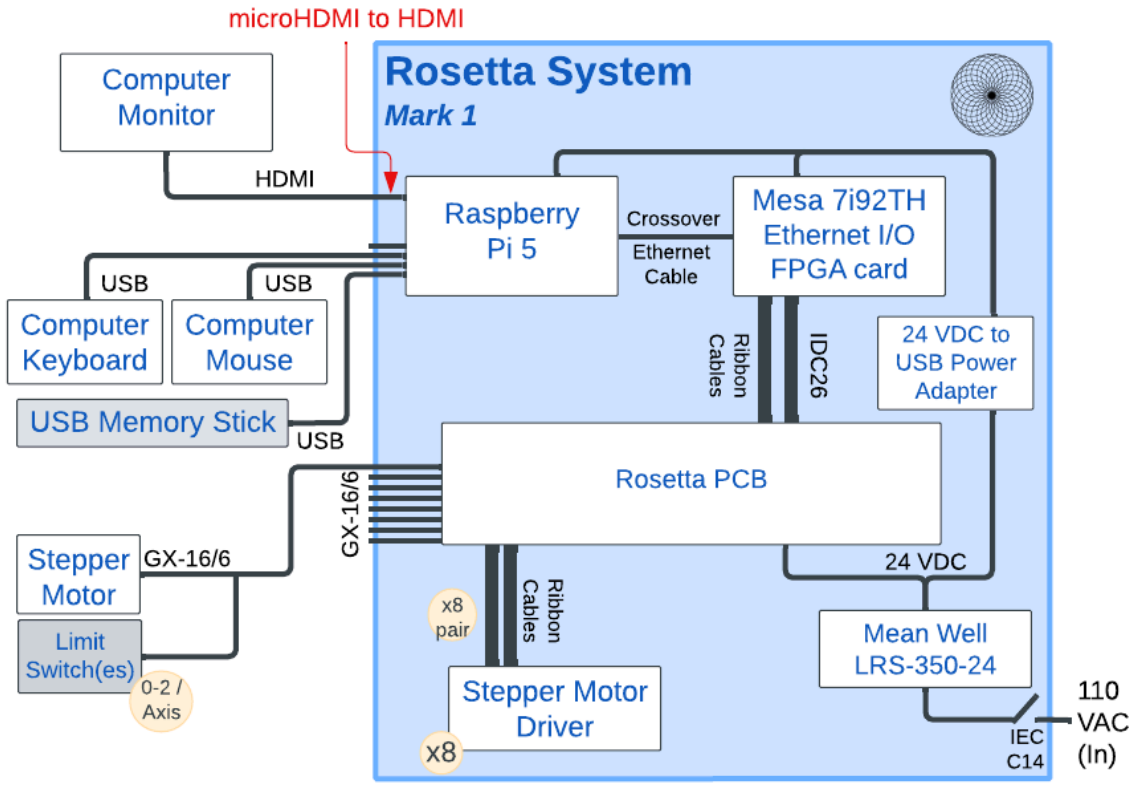
1. Computer connected to the internet with Windows, macOS, or Linux.
2. microSD card reader/writer on that computer.
3. Authority on that computer to install Raspberry Pi Imager

For running REB, you will need:

#	Item	Qty	Source	Source P/N
1000	**Case**			
1001	Case			
1100	**Power**			
1101	Power Switch	1	Astrodyne TDI	082M.01001.00-LS
1102	Power Cord	1	Schurter	6009.1214
1103	Power Outfeed Jack	1	Schurter	6600.4315
1111	24 VDC Power Supply	1		MeanWell LRS 100-24
1121	110 VAC power cable, #14			
1122	24 VDC power cable, #26			
1200	**Electronics Inside the Box**			
1201	Raspberry Pi 5 ● 8 GB RAM ● 128 GB microSD card	1	CanaKit	PI5-8GB-STR128-C4-BLK

#	Item	Qty	Source	Source P/N
1210	Mesa 7i92TH Anything I/O Ethernet card	1	Mesa US	7i92TH
1220	Stepper Motor Drivers	8	StepperOnline	DM542T
1230	REB PCB	1		
1300	**Panel Mount Parts**			
1400	**Cables within the Box**			
	IDC26-IDC26 Ribon Cable	2		
	IDC6-IDC6 Ribbon Cable	16		
	Ethernet cable	1		
1900	**Other**			
1901	Computer Monitor	1	LG	24MR400-B.AUSQ
1902	Keyboard & Mouse	1	Kensington	K72436AM

The overall connections for this system are as shown below. The items in the blue box are shipped as the REB system.

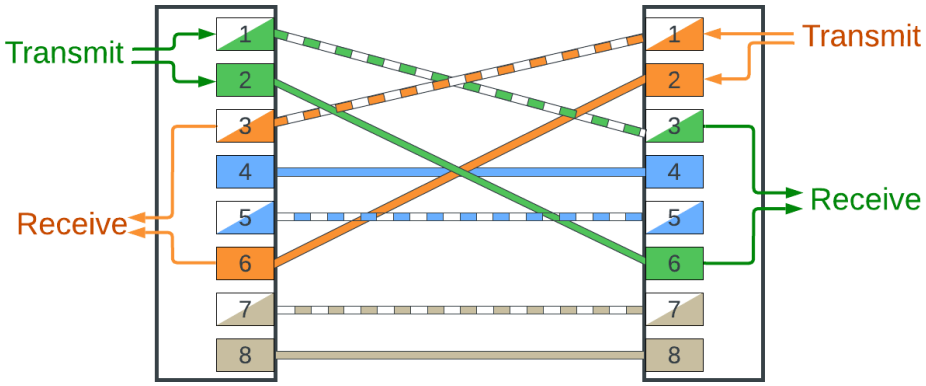


The boxes in gray and the associated cabling are optional.

The wiring configuration for a 100Mbit Ethernet crossover cable is shown to the right.

Pins 4,5, 7, & 8 still need to be wired to make this work.

There is no benefit to using a gigabit crossover cable.



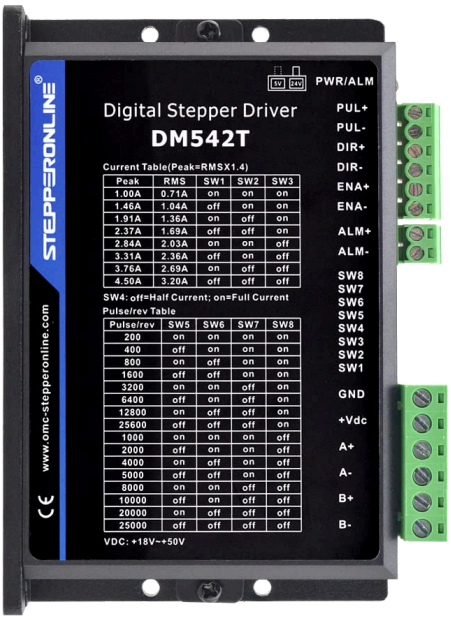
Instructions

Part 1 - Configure the Hardware

1.1 DM542T Drivers

Set the voltage to 4V (move the switch to the left as shown in the picture). Set the DIP switches on the DM542T as

Peak Current	RMS Current	SW1	SW2	SW3
1.00A	0.71A	ON	ON	ON
1.46A	1.04A	OFF	ON	ON
1.91A	1.36A	ON	OFF	ON
2.37A	1.69A	OFF	OFF	ON
2.84A	2.03A	ON	ON	OFF
3.31A	2.36A	OFF	ON	OFF
3.76A	2.69A	ON	OFF	OFF
4.50A	3.2A	OFF	OFF	OFF

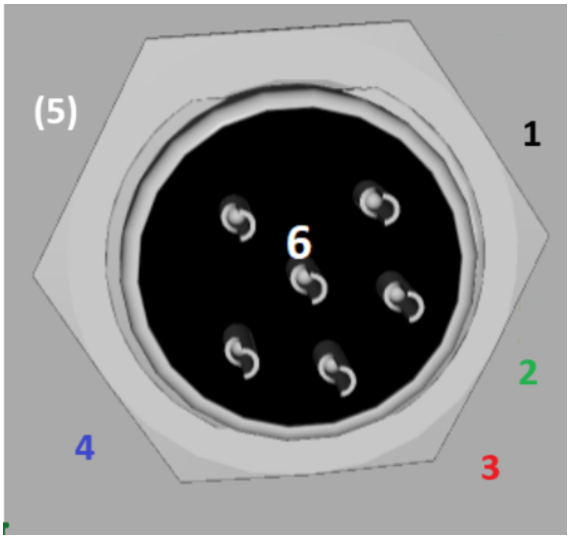


Idle current %	SW4
90%	ON
50%	OFF

Micro-steps	Steps/rev on 1.8° motor	SW5	SW6	SW7	SW8
16	3,200	ON	ON	OFF	ON
32	6,400	OFF	ON	OFF	ON
64	12,800	ON	OFF	OFF	ON
128	25,600	OFF	OFF	OFF	ON

1.2: Stepper Motors

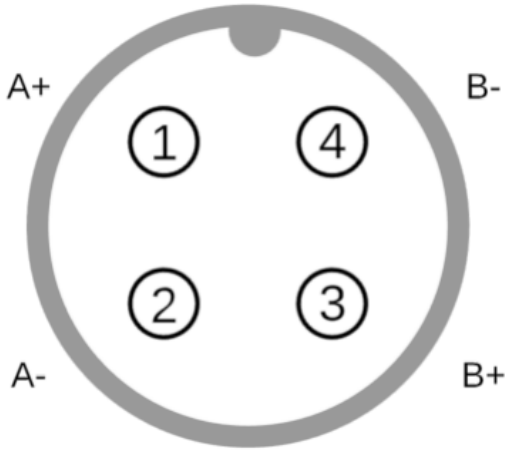
Stepper motors have 4 color coded wires. Rosetta uses a 6 pin GX16 connector so that a limit switch can be carried on the same connector. The GX16 can be mounted in a 3-D printed box that mounts to the back of the motor. This is the pinout of the GX-16 on the motor end of the cable.



Pin on GX16-6 for Motor	Signal
1	A+
2	A-
3	B+
4	B-
5	LIM
6	GND

The limit switch is shorted to ground to close the switch.

A GX16-4 connector uses only the first four wires, in the same order.



Part 2 - Build the Boot Drive for LinuxCNC

There are several distinct paths for setting up LinuxCNC on REB. You may have access to a system that is already completely set up. Or you may install a pre-configured REB installation onto an SD card. This document currently focus on the most bare option which begins with a standard LinuxCNC image and then customizes it for REB.

2.1 Get the Raspberry Pi Imager

Download and install the latest Raspberry Pi Imager (version 2.0.1 or later). This is available at

https://www.raspberrypi.com/software/

The Raspberry Pi Imager is used to create the boot disk for the Raspberry Pi using the image downloaded in the next step.

2.2 Download the LinuxCNC image file

Download this file

https://github.com/ozzyrob/linuxcnc-rpi-2.9.7-image-issues?tab=readme-ov-file

This is possibly not the same as the one available at http://linuxcnc.org/downloads.

Be sure to get the one for the Raspberry Pi 5. It is probably not the 1st one on the list.

KEY NOTE: When the system gets upgraded in step 3.3, it will upgrade the LinuxCNC to the latest version.

2.3 Create the boot disk for the Raspberry Pi

Use the Raspberry Pi Imager downloaded onto the computer from step 1 above.

a. Insert the microSD card into the computer, and ensure the computer sees it. The card will be completely overwritten. A 16G card is the smallest acceptable. Larger is desirable.

b. Start the **Raspberry Pi Imager**. For the settings, use:

Device	Raspberry Pi 5
OS	Use custom (the bottom selection) Select the file you just downloaded in step 2.
Storage	This is for the microSD card you just added in step a, above. Be sure to not select your computer's main hard drive.
Writing	This will show a summary.

c. Click **WRITE**

An alternative is to use usbimager to copy an existing SD card.

Part 3 - Configure Linux on the Raspberry Pi

A Word about Network Connections

The Raspberry Pi supports both wired and wireless Ethernet connections. For best performance, LinuxCNC prefers to have a direct wired Ethernet connection to the Mesa CNC card. Once the system is setup, the Raspberry Pi may certainly be operated without any wireless network connections. While setting up REB we assume that you connect the wired Ethernet of the Raspberry Pi to a live connection. We use this to set up the system to the point where we can switch the wired connection to access the Mesa card directly. This point is clearly delineated in the setup scripts with decisions described.

3.1 Start the Raspberry Pi

Install the microSD card into the Raspberry Pi and start the Raspberry Pi. When presented with the boot-up screen, continue.

3.2 Set System Defaults

After the system starts, a screen will popup. Before doing anything else, open a terminal emulator, and enter this command:

```
sudo passwd rpi-first-boot-wizard
```

Set the password to **chuck**. You may not need this, but if the screen locks before everything is changed over to the new username, you may need this.

In the pop-up window which came at start-up, select the values as outlined below:

Country	United States
Language	American English
Timezone	(whichever wish)

☐ Use Engish Language ☐ Use US keyboard

Click **Next**, and then enter this information.

Username	reuben
Password	chuck
Confirm Password	chuck

Click **Next**.

Make no changes on that page. Click **Next**.

Select the WiFi to use, and enter your WiFi password. Click **Next**.

Click **Next** to take the option to update the software.

Click **Launch**. The system will reboot at the end.

3.3 Set Wired Network Connection & Test Network Connections

Open a terminal emulator, and enter this command (it is a single command line):

```
nmcli con add type ethernet con-name “Wired Ethernet” ipv4.method manual ipv4.address 192.168.1.120/24
```

KEY NOTE: If you add a gateway and DNS to the wired connection, the WiFi connection will not work.

Test that everything works.

3.3.1 Test 1

Open a web browser and ensure you can open some web page, e.g., Google.com. If the wired connection is set up incorrectly, this will not work.

Close the web browser.

3.3.2 Test 2

Open a terminal emulator, and ping the Mesa 7i92TH FPGA board by entering this command:

```
ping 192.168.1.121
```

Close the terminal emulator.

3.4 Set Rest of System Options

Open a terminal emulator, and enter this command:

```
sudo raspi-config
```

Select

1 System Options

S4 Hostname
Enter **roseenginebutler**

S6 Auto Login
Select **Yes** for the question, “Would you like to automatically login to the console?”
Select **Ok**

Select **Yes** for the question, “Would you like to automatically login to the desktop?”
Select **Ok**

Select

2 Display Options

D2 Screen Blanking
Select **No** for the question, “Would you like to enable screen blanking?”
Select **Ok**

Select

Finish

Select **Yes** for the question, “Would you like to reboot now?”

3.5 Download the files needed

Open a terminal session, and enter these commands:

```
mkdir linuxcnc
cd linuxcnc
mkdir configs
cd configs
git clone https://github.com/RichColvin/RoseEngineButler
```

Once the files have been downloaded, they will be in

/home/reuben/linuxcnc/configs/RoseEngineButler

In the terminal session, enter these commands:

```
cd /home/reuben
sudo cp /home/reuben/linuxcnc/configs/RoseEngineButler/.axisrc .
```

3.6 Clean up the desktop

Right-click on the desktop. Select **Desktop Settings …**

Select the **Desktop Icons** tab

Set the icon size to 100

Select font size, and set it to 18

Select the **File/Launcher Icons** tab

Deselect all the default icons.

Part 4 - Setup LinuxCNC

4.1 Run the automated install

Ensure the Mesa 7i92TH board is

1. powered on
2. connected via Ethernet to the Raspberry Pi
    a. via an Ethernet switch or
    b. via an Ethernet crossover cable

Open a terminal emulator, and enter these commands:

```
cd /home/reuben/linuxcnc/configs/RoseEngineButler/REB_Setup
sudo bash REB_Install_Trixie.sh
```

Note: If the Mesa card has a different IP address than 192.168.1.121, replace the value used in this command with the one on your card.

```
cd /usr/lib/firmware/hm2/hostmot2
        mesaflash --device 7i92t --addr 192.168.1.121 --write 7i92t_REB.bin
        --reload
```

4.2 Set the icon on the desktop

I was not able to make the pre-defined launchers executable.

Right click on the desktop and select **Create Launcher…**. For the entries, enter:

Name	Rose Engine Butler
Comment	Desktop launcher for Rose Engine Butler
Command	linuxcnc /home/reuben/linuxcnc/configs/RoseEngineButler/REB.ini
Working Directory	/home/reuben/linuxcnc/configs/RoseEngineButler
Icon	Click the No icon button, then Select icon from image files Select /home/reuben/linuxcnc/configs/RoseEngineButler /REB_Setup/Images/ReubenHead.png

Click **Create**.

4.3 Set the wallpaper

Open the file manager. From the folder

/home/reuben/linuxcnc/configs/RoseEngineButler/REB_Setup/Images

right-click on

REBWallpaper.png

then select "**Set as wallpaper**".

4.4 Set Rose Engine Butler to auto-start when starting up the system

This is an optional step.

Click on **Applications**, then **Settings**, and then **Session and Startup**.

Click on the **Application Autostart** tab.

Click **+Add**.

For the entries, enter:

Name	Rose Engine Butler
Description	
Command	linuxcnc /home/reuben/linuxcnc/configs/REB/REB.ini
Trigger	on login

Click **OK**

4.5 Install Antivirus Application

4.5.1 Install the ClamAV application

The should have been done by the Install script. If not, these are the steps.

This is a good practice in case the user decides to keep the system connected to their Wi-Fi.

Open a terminal emulator, and enter these commands:

```
sudo apt install clamav

cd /home/reuben
mkdir ClamAv
cd ClamAv
mkdir SuspiciousFiles

sudo freshclam
```

4.5.2 Schedule the ClamAV application to run at startup

Click on **Applications**, then **Settings**, and then **Session and Startup**.

Click on the **Application Autostart** tab.

Click **+Add**.

For the entries, enter:

Name	ClamAV
Description	Antivirus software
Command	sudo clamscan -r -i / --move=/home/reuben/ClamAV/SuspiciousFiles --log=/home/reuben/ClamAV/ScanLog.txt
Trigger	on login

Click **OK**

4.5.3 Schedule the ClamAV application to run weekly

This will schedule the virus scan to run each Sunday morning at 03:00. This will take place if the user does not shutdown and leaves the system running.

Open a terminal emulator, and enter this command:

```
crontab -e
```

Choose option 1. On the bottom of the file, enter these 2 lines (each is a single line):

0 1 * * 1 sudo freshclam

0 3 * * 1 sudo clamscan -r -i / --move=/home/reuben/ClamAV/SuspiciousFiles --log=/home/reuben/ClamAV/ScanLog.txt

Click **Cntrl-O** to save the file, and then **Cntrl-X** to exit.

4.6 Install XFCE Menu Maintenance Application

4.6.1 Install the menulibre application

This program is used to manipulate the XFCE Applications menu

Open a terminal emulator, and enter this command:

```
sudo apt install menulibre
```

4.6.2 Run the menulibre application

Open a terminal emulator, and enter this command (no sudo is required):

```
menulibre
```

4.6.3 Add a Rose Engine Butler menu

1. Click the large + on the upper left part of the screen.
2. Select **Add Directory**.
3. Change “New Directory” to “Rose Engine Butler”
4. Change “A small descriptive blurb about this directory” to “This is for the Rose Engine Butler programs.”
5. Click on the folder icon, and replace it with ReubenHead.png from the directory, */home/reuben/linuxcnc/configs/RoseEngineButler/REB_Setup/Images/*.
6. Click on the save button (the one beside the large + on the upper left part of the screen).
7. Using the Up arrow at the bottom of the screen on the left, and move this menu to be above **Terminal Emulator**.

4.6.4 Add the Rose Engine Butler update program

1. Click the large + on the upper left part of the screen.
2. Select **Add Launcher**.

3. Change “New Launcher” to “Update Rose Engine Butler”
4. Change “A small descriptive blurb about this application” to “Run this when notified to do so”.
5. Click on the icon, and replace it with an up arrow icon.
6. For **Command**, enter “bash /home/reuben/linuxcnc/configs/REB_Update.sh”
7. For **Working Directory**, enter “/home/reuben/linuxcnc/configs/RoseEngineButler/”
8. Under **Options**, enable these two options:
    a. Run in terminal
    b. Use startup notification
9. In the **Categories** box, delete **X-Xfce-Toplevel**.
10. Click on the save button (the one beside the large + on the upper left part of the screen).

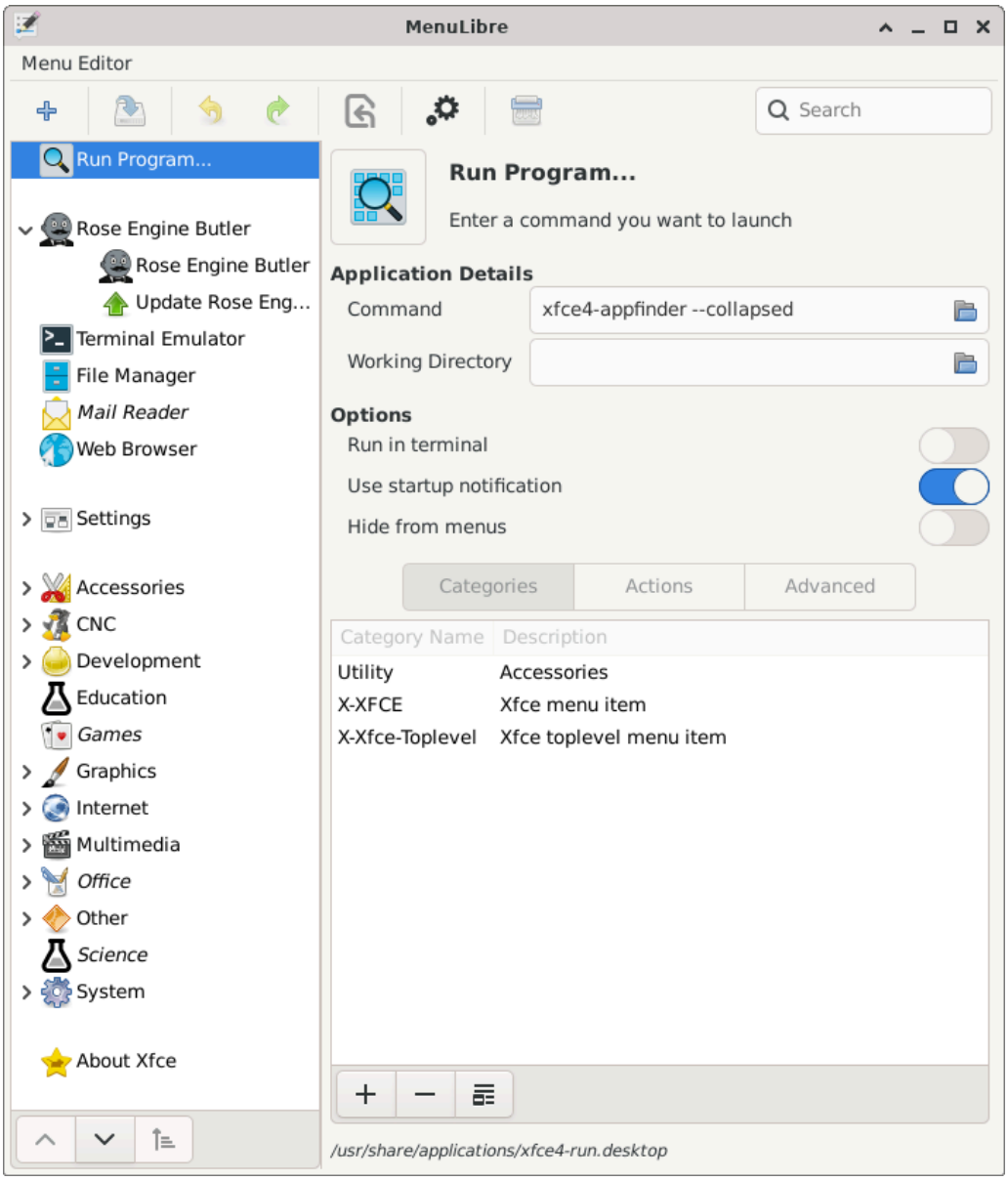
4.6.5 Add the Rose Engine Butler program

1. Click the large + on the upper left part of the screen.
2. Select **Add Launcher**.
3. Change “New Launcher” to “Rose Engine Butler”
4. Change “A small descriptive blurb about this application” to “Desktop launcher for the Rose Engine Butler”
5. Click on the icon, and replace it with ReubenHead.png from the directory, */home/reuben/linuxcnc/configs/RoseEngineButler/REB_Setup/Images/*.
6. For **Command**, enter “linuxcnc /home/reuben/linuxcnc/configs/RoseEngineButler/REB.ini”
7. For **Working Directory**, enter “/home/reuben/linuxcnc/configs/RoseEngineButler/”
8. In the **Categories** box, delete **X-Xfce-Toplevel**.
9. Click on the save button (the one beside the large + on the upper left part of the screen).

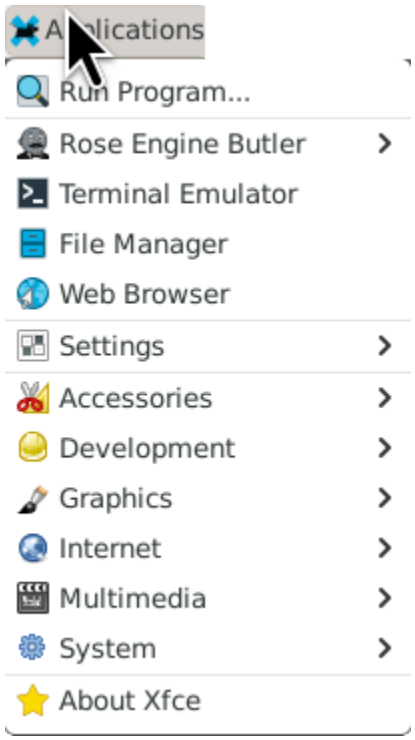
4.6.6 Hide some unnecessary programs / program menus

1. For each of these applications, under Options, enable the option **Hide from menus**, saving each after the change
    a. Games
    b. Mail Reader
    c. Office
    d. Science
2. Click on the save button (the one beside the large + on the upper left part of the screen).

The resulting MenuLibre screen should look like the picture below:



Which results in the Applications menu shown below.



Part 5 - Install Development System Applications

Install these applications on a development system. They are not needed on production systems, but can be installed if desired.

5.1 Glade

Glade is used to develop the display panels used with GladeVCP.

Open a terminal emulator, and enter these commands:

```
sudo apt install glade
```

5.2 xfce4-screenschooter

xfce4-screenshooter is used for screen captures. It is quite useful for documentation purposes.

Open a terminal emulator, and enter these commands:

```
sudo apt install xfce4-screenshooter
```

To use it, press the PrtScn button, and then follow the prompts. The images are stored in the Pictures directory.

5.3 GIMP

The GNU Image Manipulation Program is useful for editing image files.

Open a terminal emulator, and enter these commands:

```
sudo apt install gimp
```

Reference Information

Reference information is provided in the support manual (support.RoseEngineButler.com).

Modification History

Version	Date	Comments
1.0		

Optional Notes

To install the Chromium browser:

- echo "deb http://deb.debian.org/debian bookworm main contrib non-free non-free-firmware" | sudo tee /etc/apt/sources.list.d/debian.list
- sudo apt update
- sudo apt install chromium -y

To SSH in:
ssh cnc@192.168.1.12 (use the IP address of your Pi)

To unzip:
sudo apt update
sudo apt install file-roller

WiFi
LinuxCNC takes over the Ethernet port when it runs, so it's best to use a crossover cable and not rely on this for external access. Bring up WiFi if you want to have an Ethernet connecton.
It took me a bit of effort to get WiFi working using 6.12.11.

- sudo apt install network-manager
- sudo nmcli connection add type wifi ifname wlan0 con-name MyWiFi ssid "yellowcat"
- sudo nmcli connection modify MyWiFi wifi-sec.key-mgmt wpa-psk
- sudo nmcli connection modify MyWiFi wifi-sec.psk "home4349"
- sudo nmcli connection up MyWiFi
- nmcli device status
- nmcli connection modify wlan0 ipv4.route-metric 100
- nmcli connection modify eth0 ipv4.route-metric 200
- sudo nmcli device set wlan0 managed yes
- sudo systemctl restart NetworkManager
- sudo systemctl status wpa_supplicant
- sudo rm /run/wpa_supplicant/wlan0
- sudo nmcli device set wlan0 managed yes
- sudo systemctl restart NetworkManager